Embedded Linux – Chapter 1: Introduction

Based on Embedded Linux Primer (2nd Edition)

Course: Embedded Linux Systems

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Learning Objectives

- Understand why Linux is widely used in embedded systems
- Explain current trends of Embedded Linux
- Understand Open Source & GPL concepts
- Identify standards and organizations in Embedded Linux

What is Embedded Linux?

- Linux OS used in embedded systems
- Embedded systems are designed for specific tasks
- Often resource-constrained (CPU, memory, power)
- Examples: smartphones, smart TVs, routers, automotive systems

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Embedded Systems in Real Life



Why Linux for Embedded Systems?

- Supports many hardware architectures
- Supports many applications & networking protocols
- Highly scalable
- No royalty cost compared to proprietary OS
- Strong developer community and vendor support

Key Advantage: Scalability

- Runs on small devices (IoT, sensors)
- Runs on medium devices (routers, TVs)
- Runs on large systems (servers, telecom)
- One OS for multiple platforms

Embedded Linux Today

- Linux is dominant in embedded systems
- ~50% of engineers use Linux in designs
- Traditional OS usage is decreasing
- Rapid growth trend continues

Open Source Concept

- Source code is publicly available
- Developers can study, modify, and distribute
- Encourages collaboration and innovation
- Open source does not always mean free of cost
- Free vs Freedom

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GPL (General Public License)

- Freedom to run the software
- Freedom to study and modify source code
- Freedom to redistribute
- Modified code must remain open (viral license)
- GPL v2 and GPL v3

Hidden Cost of Linux

- Development and integration cost
- Maintenance and support cost
- Toolchain and environment setup
- Linux is free to download, not free to develop

Standards in Embedded Linux

- Linux Standard Base (LSB)
- Defines libraries, file system layout, commands
- Ensures cross-distribution compatibility

Linux Foundation

- Non-profit organization
- Supports Linux ecosystem
- Sponsors kernel development
- Encourages industry collaboration

Carrier-Grade Linux (CGL)

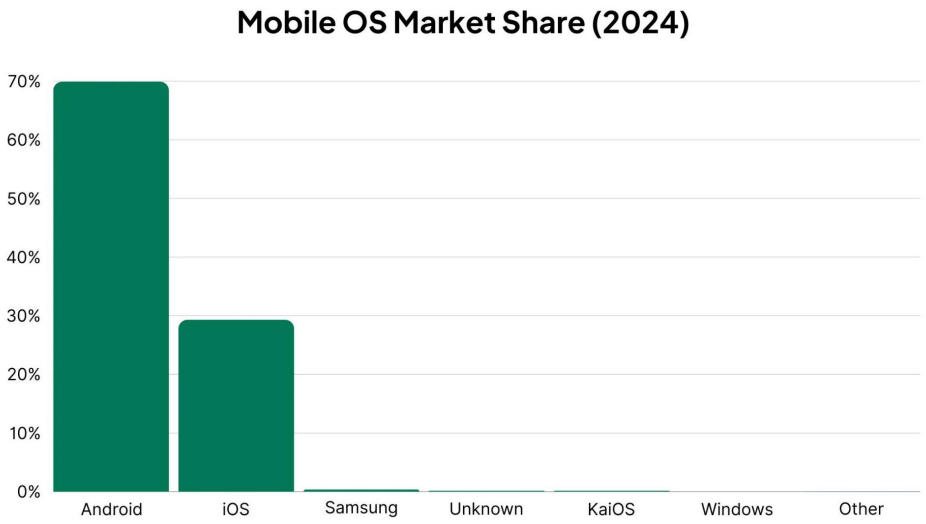
- Used in telecom and network systems
- High availability and reliability
- Security and performance requirements
- Supports clustering and fault tolerance

→ In carrier-grade systems, Linux is not used as-is — it becomes a heavily customized platform, and the cost shifts from licensing to engineering and long-term maintenance

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Mobile Linux → Android

- Used in smartphones and portable devices
- Projects: Moblin, MeeGo (death)
- Focus on power management, multimedia, security



Service Availability Forum

- Focus on high reliability systems (RAS)
- Defines standard interfaces
- Used in telecom and enterprise systems

Summary

- Linux is widely used in embedded systems
- Driven by flexibility and cost advantage
- Open Source and GPL are key concepts
- Standards and organizations support ecosystem

Discussion Questions

- Why do companies prefer Linux over proprietary OS?
- What are risks of using GPL software?
- Is Linux truly free?

Suggested Reading

- The Cathedral and the Bazaar – Eric Raymond
- Linux Foundation website
- GPL License documentation